mitsue_wbs

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Mitsue Project — Work Breakdown Structure

Phases 0–3 · April 2026 – September 2028

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1. How to Read This Document

The WBS decomposes the entire project scope into discrete, manageable elements. Each element at the **lowest level** (the work package) represents a bounded unit of work with a defined deliverable, budget, and responsible party.

Conventions used: - WBS codes use dot notation: `1.0` = Level 1, `1.2` = Level 2, `1.2.3` = Level 3 - **Budget** columns show the planning estimate; ranges indicate remaining uncertainty - **Owner** is the person accountable for delivery (not necessarily the doer) - **Deliverable** is what must exist / be accepted for the element to be marked complete - Total BAC = **¥220.0M** | Management Reserve (not in WBS) = ¥25.0M - **Baseline Rev 1 (May 2026)** — Reality-checked against real-world Japan benchmarks for solar PV (¥200–300K/kW), school seismic retrofit (¥200–500K/㎡), commercial EV fast-chargers (¥5–6M each), and forestry road works. BAC raised from ¥168M to ¥220M.

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2. WBS Summary Tree

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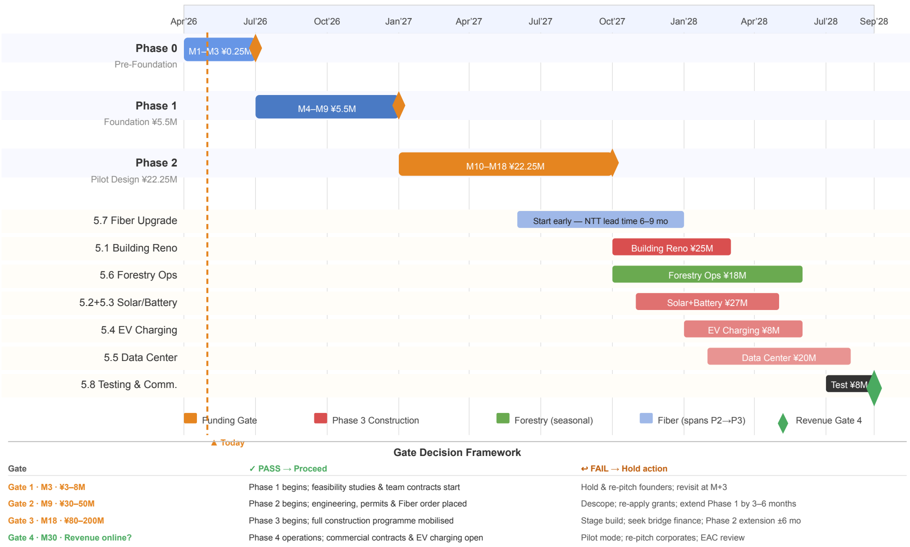
0.0  MITSUE PROJECT (Phases 0–3)                          ¥220.0M
 │
 ├── 1.0  Project Management & Governance                  ¥15.0M
 │     ├── 1.1  Programme direction & reporting
 │     ├── 1.2  Legal, accounting & compliance
 │     └── 1.3  Communications, translation & website
 │
 ├── 2.0  Phase 0 — Pre-Foundation                          ¥0.25M
 │     ├── 2.1  Community stakeholder engagement
 │     ├── 2.2  Founding charter & documents
 │     └── 2.3  Founding team identification
 │
 ├── 3.0  Phase 1 — Foundation                              ¥5.5M
 │     ├── 3.1  Legal entity setup (一般社団法人)
 │     ├── 3.2  Forestry feasibility study
 │     ├── 3.3  Energy systems feasibility study
 │     ├── 3.4  Building & site assessment
 │     ├── 3.5  Connectivity assessment
 │     └── 3.6  Operations setup
 │
 ├── 4.0  Phase 2 — Pilot Design                           ¥22.25M
 │     ├── 4.1  Building & structural design
 │     ├── 4.2  Energy systems engineering
 │     ├── 4.3  Data center & IT architecture
 │     ├── 4.4  EV charging system design
 │     ├── 4.5  Permitting & regulatory
 │     ├── 4.6  Partnership & landowner agreements
 │     ├── 4.7  Grant writing & funding applications
 │     ├── 4.8  Staff recruitment & onboarding
```

```
│       ├── 4.9  Vendor pre-qualification
│       └── 4.10 Phase 2 contingency
│
└── 5.0  Phase 3 — Pilot Build                           ¥177.0M
        ├── 5.1  School building renovation
        ├── 5.2  Solar PV installation
        ├── 5.3  Battery storage system
        ├── 5.4  EV charging infrastructure
        ├── 5.5  Data center fitout
        ├── 5.6  Forestry operations
        ├── 5.7  Fiber connectivity upgrade
        ├── 5.8  Testing, commissioning & startup
        └── 5.9  Phase 3 contingency
```

Figure 1 — Project Phase Timeline (Gantt View, Apr 2026 – Sep 2028)



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3. WBS Dictionary — Full Detail

1.0 Project Management & Governance — ¥15.0M

Cross-phase overhead covering leadership, legal compliance, and external communications. These costs are time-phased across all 30 months and do not correspond to a single phase.

1.1 Programme Direction & Reporting — ¥8.0M

Item	Detail
Scope	Representative Director oversight; monthly EVM reporting; board meetings; stakeholder briefings; risk register maintenance
Owner	Representative Director (代表理事)
Budget	¥8.0M (¥265K/month average across 30 months)
Deliverables	Monthly EVM status reports; gate review packages; annual stakeholder reports
Acceptance	Board approval of each gate review package

Work packages: - Monthly EVM status reports (M1–M30) - Quarterly stakeholder briefings (8 reports) - Four gate review packages (Gate 1–4) - Annual year-end reviews (2026, 2027, 2028) - Risk register (maintained monthly) - OpenProject administration

1.2 Legal, Accounting & Compliance — ¥4.0M

Item	Detail
Scope	行政書士 retainer for grants & permits; 税理士 for annual tax filing; legal counsel for contracts; ongoing compliance
Owner	Director — Finance/Operations
Budget	¥4.0M
Deliverables	Audited annual accounts; tax filings; legal review sign-offs on all major contracts
Acceptance	Annual accounts accepted by auditor (監事)

Work packages: - 行政書士 retainer (Phase 1–3) - 税理士 / 公認会計士 annual engagement - Annual financial report & audit - Contract legal review (landowner, vendor, partnership agreements)

1.3 Communications, Translation & Website — ¥3.0M

Item	Detail
Scope	Bilingual project website; press & stakeholder materials; translation of all formal documents; annual report design
Owner	Director — Technology
Budget	¥3.0M
Deliverables	Live bilingual website (by end of Phase 1); translated versions of all key documents; annual impact report
Acceptance	Website live and reviewed by board before Phase 1 close

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2.0 Phase 0 — Pre-Foundation — ¥0.25M

Duration: Month 1–3 (April–June 2026)

Low-budget, high-importance phase. The primary currency is time and trust, not money. Goal is to establish social license before any public commitments are made.

2.1 Community Stakeholder Engagement — ¥0.10M

Item	Detail
Scope	Private meetings with village mayor, council members, 自治会 leaders in Sugano and nearby hamlets; listening sessions (not pitching)
Owner	Rob Oudendijk + Japanese co-founder (when identified)
Budget	¥0.10M (travel, hospitality, materials)
Deliverables	Meeting log with attendees and key reactions; summary memo of village leadership posture (supportive / neutral / cautious)
Acceptance	Village mayor informed; no active opposition identified

2.2 Founding Charter & Documents — ¥0.10M

Item	Detail
Scope	Draft 2-page bilingual charter; preliminary project description in Japanese; translation of core concept documents
Owner	Rob Oudendijk
Budget	¥0.10M (translation fees, 行政書士 preliminary consultation)
Deliverables	Draft charter (日本語・英語); translated project concept note
Acceptance	Charter reviewed and agreed by founding team members

2.3 Founding Team Identification — ¥0.05M

Item	Detail
Scope	Identify and secure verbal commitments from 3–5 founding team members; at minimum one well-respected Japanese co-founder with rural credibility
Owner	Rob Oudendijk
Budget	¥0.05M (networking, travel)
Deliverables	Named list of founding team with verbal commitments documented
Acceptance	Japanese Representative Director candidate identified

> **Phase 0 Gate Criterion:** Founding team agreed (verbal); village leadership informed and not opposed; draft charter complete.

3.0 Phase 1 — Foundation — ¥5.5M

Duration: Month 4–9 (July–December 2026)

The most structurally important phase. The feasibility studies produced here determine the credibility of all subsequent funding applications and Phase 2/3 budgets. Cutting corners here is false economy.

3.1 Legal Entity Setup — ¥0.5M

Item	Detail
Scope	一般社団法人 incorporation; notarized 定款; Legal Affairs Bureau (法務局) registration; opening bank account; accounting system setup (弥生会計 or equivalent)
Owner	Director — Finance/Operations
Budget	¥0.5M (¥110K registration + ¥50K notary + ¥300K scrivener + ¥40K accounting setup)
Deliverables	Certificate of incorporation; registered office address; operational bank account; accounting system active
Acceptance	法務局 registration confirmed; first board resolution recorded

3.2 Forestry Feasibility Study — ¥1.5M

Item	Detail
Scope	Field survey of candidate sugi plots; thinning residue yield estimates (fuel input for biomass-electric conversion); transportation cost analysis from steep terrain; native forest restoration plan and timeline; carbon sequestration estimate; coordination with 日本林業協会 / 林野庁 advisors
Owner	Director — Forestry/Local
Budget	¥1.5M (lower end of ¥1.5–3M range; full range covered by Phase 1 contingency if needed)
Deliverables	Forestry feasibility report (Japanese); executive summary (bilingual); land parcel map; carbon estimate methodology
Acceptance	Report accepted by advisory board; J-Credit pre-qualification confirmed or ruled out

3.3 Energy Systems Feasibility Study — ¥1.5M

Item	Detail
Scope	Solar generation potential at school site; battery storage sizing; grid connection options; FIT/FIP eligibility assessment; EV charging load modelling; blackout resilience scenarios
Owner	Director — Technology
Budget	¥1.5M (lower end of ¥2–4M range)
Deliverables	Energy feasibility report; solar yield model; battery sizing recommendation; grid connection options memo; FIT/FIP eligibility letter
Acceptance	Report accepted by advisory board; METI pre-consultation completed

3.4 Building & Site Assessment — ¥0.8M

Item	Detail
Scope	Structural condition survey of Mitsue Elementary School; seismic compliance assessment (新耐震基準); identification of required renovations for data center use; zoning review
Owner	Director — Technology
Budget	¥0.8M (mid of ¥1–2M range for single-building survey)
Deliverables	Structural survey report; seismic compliance summary; renovation scope outline; zoning confirmation memo
Acceptance	Report accepted by board; Phase 3 renovation budget range confirmed

3.5 Connectivity Assessment — ¥0.4M

Item	Detail
Scope	Current fiber capacity reaching Mitsue; upgrade requirements and costs; NTT consultation; satellite/microwave fallback options
Owner	Director — Technology
Budget	¥0.4M (mid of ¥0.3–0.5M range)
Deliverables	Connectivity gap analysis; NTT meeting notes; upgrade cost estimate; backup options memo
Acceptance	Report accepted; Phase 3 fiber upgrade budget confirmed

3.6 Operations Setup — ¥0.8M

Item	Detail
Scope	Advisory board formalization (letters of commitment from Joi Ito, Ray Ozzie, others); initial bilingual website live; funding pipeline documentation; grant calendar setup
Owner	Representative Director
Budget	¥0.8M
Deliverables	Signed advisory board letters; live website; grant application calendar for Phase 2
Acceptance	Website reviewed by board; advisory board confirmed in writing

> **Phase 1 Gate Criterion:** Legal entity registered; all four feasibility studies accepted; ¥3–8M secured; letter of support from Mitsue village government.

4.0 Phase 2 — Pilot Design — ¥22.25M

Duration: Month 10–18 (January–September 2027)

Converts feasibility into engineering reality. The permit applications submitted here are the long-lead items that drive Phase 3 start. Staffing up in this phase is critical.

4.1 Building & Structural Design — ¥2.0M

Item	Detail
Scope	Full architectural drawings for school renovation (1 wing); seismic retrofit specifications; mechanical/electrical/plumbing for data center wing; building permit application package
Owner	Director — Technology
Budget	¥2.0M
Deliverables	Stamped architectural drawings; MEP specs; building permit application submitted
Acceptance	Building permit application accepted by 奈良県 authorities

4.2 Energy Systems Engineering — ¥2.0M

Item	Detail
Scope	Detailed design for solar PV array, battery storage, EV charging integration, and grid connection; equipment specifications; load flow analysis; FIT/FIP application
Owner	Director — Technology
Budget	¥2.0M
Deliverables	Detailed energy system design package; equipment spec sheets; FIT/FIP application submitted to METI
Acceptance	METI FIT/FIP application reference number received

4.3 Data Center & IT Architecture — ¥1.0M

Item	Detail
Scope	Server room layout; power and cooling design for 10–20 servers; network architecture; APPI compliance plan; cybersecurity framework
Owner	Director — Technology
Budget	¥1.0M
Deliverables	Data center design package; network topology diagram; security framework document
Acceptance	Design reviewed and accepted by technical advisory committee

4.4 EV Charging System Design — ¥1.0M

Item	Detail
Scope	Charging station layout for 4 units; grid integration design; 消防 and 電気設備 permit preparation; signage and access plan
Owner	Director — Technology
Budget	¥1.0M

Item	Detail
Deliverables	EV charging design drawings; permit pre-application submitted
Acceptance	Fire department and electrical safety pre-consultation completed

4.5 Permitting & Regulatory — ¥3.5M

Item	Detail
Scope	Manage all permit applications across agencies: building permit (建築基準法), forestry cutting notification (伐採届出), FIT/FIP registration (METI), EV charging safety permits (消防・電気), environmental review coordination
Owner	Director — Finance/Operations (supported by 行政書士)
Budget	¥3.5M (includes 行政書士 fees of ¥200–500K per application)
Deliverables	All permit applications submitted; building permit issued; FIT/FIP registration confirmed
Acceptance	Core permits in hand before Phase 3 procurement begins

> **Note:** METI FIT/FIP registration and building permits are the critical-path items — typical lead time 4–6 months. Applications must be submitted no later than M12 (March 2027).

4.6 Partnership & Landowner Agreements — ¥1.5M

Item	Detail
Scope	Negotiate and execute landowner contracts for sugi harvesting rights; memoranda of understanding with key technical and institutional partners; village government cooperation agreement
Owner	Representative Director
Budget	¥1.5M (legal drafting, negotiation facilitation, translation)
Deliverables	Minimum 2 signed landowner harvest agreements; village government MOU signed; at least 1 corporate partner MOU
Acceptance	Legal review sign-off on all agreements; board approval

4.7 Grant Writing & Funding Applications — ¥2.0M

Item	Detail
Scope	Prepare and submit applications for major grants: 地方創生関係交付金, 林野庁 subsidies, NEDO, METI green tech, Nara Prefecture, Nippon Foundation, Japan Fund for Global Environment
Owner	Director — Finance/Operations (supported by specialist 行政書士)
Budget	¥2.0M (行政書士 fees ¥200–500K each; 4–6 applications)

Item	Detail
Deliverables	Minimum 4 grant applications submitted; at least ¥30M in pending applications by end of Phase 2
Acceptance	Applications submitted; Gate 3 funding of ¥80–200M confirmed or committed

4.8 Staff Recruitment & Onboarding — ¥6.0M

Item	Detail
Scope	Hire first 2–3 part-time staff (project coordinator, operations/admin, possibly local forestry liaison); onboarding and process documentation
Owner	Representative Director
Budget	¥6.0M (9 months × 3 × ~¥220K/month blended part-time)
Deliverables	2–3 staff under contract; onboarding complete; key processes documented
Acceptance	Staff in post and contributing before Phase 3 begins

4.9 Vendor Pre-Qualification — ¥1.5M

Item	Detail
Scope	Shortlist and pre-qualify vendors for: building renovation contractor, solar/battery supplier, data center hardware, EV charging equipment; obtain binding budget quotes for Phase 3 procurement
Owner	Director — Technology
Budget	¥1.5M
Deliverables	Vendor shortlist per category; budget quotes received; Phase 3 procurement plan
Acceptance	Phase 3 WBS 5.x budgets confirmed within ±15% of EVM baseline before Gate 3

4.10 Phase 2 Contingency — ¥1.75M

Item	Detail
Scope	Reserve for Phase 2 cost variances — particularly permit fees higher than estimated, additional engineering iterations, or grant application costs exceeding plan
Owner	Representative Director
Budget	¥1.75M (~8% of Phase 2 direct costs)
Access	Requires Representative Director approval; drawn down only against specific identified variances

> **Phase 2 Gate Criterion:** ¥30–50M secured; detailed engineering complete; key permits in hand (or application submitted with favourable pre-consultation); 2 staff in post.

5.0 Phase 3 — Pilot Build — ¥177.0M

Duration: Month 19–30 (October 2027 – September 2028)

The largest and most complex phase. Peak spend is in Months 22–27. All Phase 3 work packages are interdependent — building renovation must precede data center fitout; solar/battery must precede EV commissioning.

Sequencing logic:

```
5.1 Building Renovation → 5.5 Data Center Fitout
5.2 Solar PV + 5.3 Battery → 5.4 EV Charging commissioning
5.6 Forestry Operations → independent track (weather/season dependent)
5.7 Fiber → early in P3 (long lead time with NTT)
5.8 Testing & Commissioning → final 2 months
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5.1 School Building Renovation — ¥38.0M

Item	Detail
Scope	Seismic retrofit of 1 wing; internal renovation for data center and office use; MEP installation (power distribution, cooling, fire suppression); accessible entrance and welfare facilities; exterior weatherproofing
Owner	Director — Technology
Budget	¥38.0M
Deliverables	Renovated wing with 建築基準法 compliance certificate; handover from contractor; punch-list sign-off
Acceptance	Building inspector sign-off; internal acceptance by project board
Key risks	Unforeseen structural issues adding scope; labour shortage in rural Nara

Work packages: - Seismic retrofit (structural steel/concrete) - MEP rough-in (power, cooling, fire suppression) - Internal partitioning and fit-out - Exterior repairs and weatherproofing - Site works (access road, parking, landscaping for EV) - Final inspections and compliance certificates

5.2 Solar PV Installation — ¥22.0M

Item	Detail
Scope	Rooftop solar PV (~100 kW); inverters; monitoring system; grid connection infrastructure; commissioning and FIT/FIP handover documentation
Owner	Director — Technology
Budget	¥22.0M (approx. ¥220K/kW installed, rural premium)
Deliverables	Grid-connected solar array generating power; FIT/FIP connection agreement in place; monitoring dashboard live
Acceptance	METI FIT/FIP connection confirmed; generation data logging active

5.3 Battery Storage System — ¥12.0M (subject to feasibility study confirmation)

Item	Detail
Scope	Lithium-ion battery storage system sized for 12–48 hours critical-facility backup; BMS; integration with solar and grid; safety certifications. **Decision gate:**

Item	Detail
	Phase 1 energy feasibility study must confirm economic and operational case before this work package proceeds.
Owner	Director — Technology
Budget	¥12.0M
Deliverables	Battery system installed, commissioned, and safety-certified; blackout simulation test passed
Acceptance	消防 safety certificate; 48-hour backup test at design load

5.4 EV Charging Infrastructure — ¥15.0M

Item	Detail
Scope	4 EV charging stations (mix of standard AC and fast DC); civil works for cable trenching; payment/management software; signage; safety approvals; public launch
Owner	Director — Technology
Budget	¥15.0M
Deliverables	4 operational charging stations open to public; payment system live; 消防・電気設備 approvals in hand
Acceptance	Public launch event; village government acceptance

5.5 Data Center Fitout — ¥20.0M

Item	Detail
Scope	Server racks and hardware (10–20 servers, edge computing focus); precision cooling; structured cabling; UPS backup integration with battery system; network equipment; APPI-compliant security controls; remote monitoring
Owner	Director — Technology
Budget	¥20.0M
Deliverables	Operational data center accepting first paying workloads; monitoring dashboard live; APPI compliance documentation complete
Acceptance	First commercial hosting contract signed; uptime monitoring active

5.6 Forestry Operations — ¥25.0M

Item	Detail
Scope	First 5–10 ha of sugi harvested under signed landowner agreements; timber processed for lumber + residue stockpiled for future biomass-electric conversion; native species replanting commenced; forest road and drainage improvements; J-Credit documentation initiated
Owner	Director — Forestry/Local
Budget	¥25.0M

Item	Detail
Deliverables	Harvest completion report; replanting plan executed for cleared area; first J-Credit application submitted
Acceptance	伐採届出 (cutting notification) filed and acknowledged; replanting confirmed by site inspection
Key risks	Seasonal access (avoid typhoon season M19–M23); steep terrain requiring specialised equipment

5.7 Fiber Connectivity Upgrade — ¥10.0M

Item	Detail
Scope	Upgrade fiber capacity to Mitsue in coordination with NTT and/or regional ISP; install dedicated dark fiber or upgraded leased line to school site; microwave backup link installation
Owner	Director — Technology
Budget	¥10.0M
Deliverables	Confirmed bandwidth upgrade (target: 1 Gbps symmetric minimum to school site); backup link live; NTT SLA in place
Acceptance	Bandwidth test confirming contracted speeds; failover test on backup link
Note	NTT coordination has 6–9 month lead time — initiate in Phase 2 (M15 at latest)

5.8 Testing, Commissioning & Startup — ¥8.0M

Item	Detail
Scope	Integrated system testing across all Phase 3 elements; punch-list resolution; staff training; documentation finalisation; public launch event; Phase 4 operational handover
Owner	Representative Director
Budget	¥8.0M
Deliverables	All systems tested and operational; staff trained; punch-list closed; Phase 4 operational plan approved; launch event completed
Acceptance	Board acceptance of Phase 3 completion; Phase 4 operational plan approved

Sub-work packages: - Solar + battery integrated test (blackout simulation) - Data center stress test and security audit - EV charging operational test (all 4 stations) - Forestry first-year progress review - Fiber latency and failover test - Staff training programme (data center ops, EV maintenance, forestry coordination) - Phase 3 completion documentation package - Public launch event (village + media + funders)

5.9 Phase 3 Contingency — ¥27.0M

Item	Detail
Scope	Reserve for Phase 3 cost variances — primarily structural findings in 5.1, equipment cost movements, rural labour premiums, or seasonal delays
Owner	Representative Director

Item	Detail
Budget	¥27.0M (~18% of Phase 3 direct costs)
Access	Representative Director approval for draws up to ¥5M; board approval required above ¥5M

> **Phase 3 Gate Criterion (Gate 4):** All systems operational; first commercial data center contract signed; EV charging open to public; forestry harvest commenced; Phase 4 plan approved by board.

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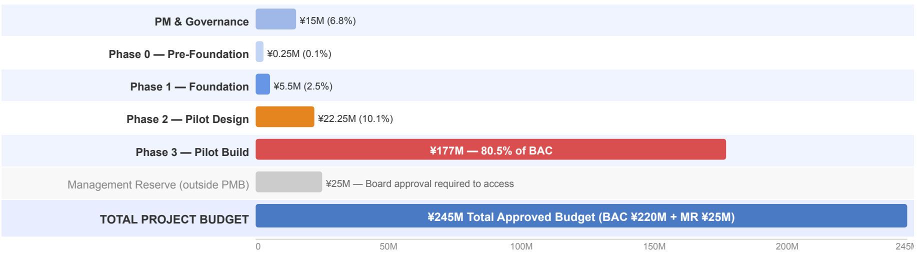
4. Budget Summary

WBS	Element	Budget (¥M)	% of BAC
1.0	Project Management & Governance	15.00	6.8%
2.0	Phase 0 — Pre-Foundation	0.25	0.1%
3.0	Phase 1 — Foundation	5.50	2.5%
4.0	Phase 2 — Pilot Design	22.25	10.1%
5.0	Phase 3 — Pilot Build	177.00	80.5%
	Total BAC (PMB)	**220.00**	**100%**
	Management Reserve (not in PMB)	25.00	—
	Total Project Budget	**245.00**	—

Figure 2 — Budget by WBS Element

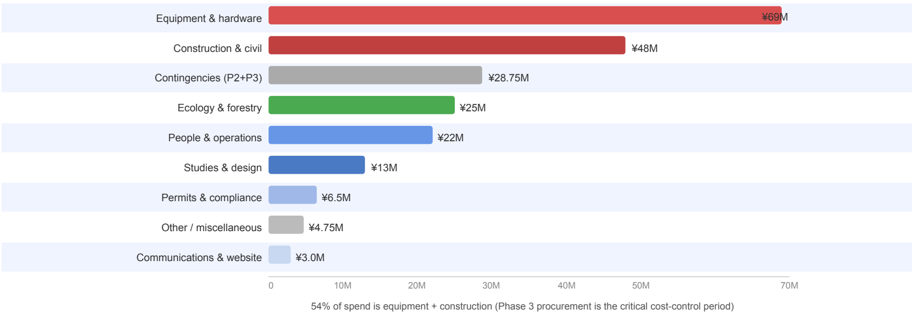


Budget by Category (across all phases)

Category	Budget (¥M)	Examples
Studies & design	¥13.0M	Feasibility studies, engineering design, IT architecture
Construction & civil	¥48.0M	Building renovation, site works, civil for EV/fiber
Equipment & hardware	¥69.0M	Solar, battery, EV chargers, servers, network gear
Ecology & forestry	¥25.0M	Harvest operations, replanting, forest roads
Permits & compliance	¥6.5M	行政書士 fees, permit applications, legal
People & operations	¥22.0M	PM team, hired staff, advisory board
Contingencies	¥28.75M	Phase 2 and Phase 3 contingency reserves

Category	Budget (¥M)	Examples
Communications	¥3.0M	Website, translation, design
Other / miscellaneous	¥4.75M	Accounting, banking, training, launch event
Total BAC	**¥220.0M**	

Figure 3 — Budget Breakdown by Cost Category



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5. Key Dependencies Map

Dependency	Predecessor	Successor	Consequence if late
Japanese co-founder	2.3	3.1, 3.6	Cannot incorporate without Rep. Director
Village govt support	2.1	3.1, 4.6	No MOU; weakens grant applications
Feasibility studies	3.2–3.5	4.1–4.4, gate funding	Phase 2 budgets unreliable; grants harder
METI FIT/FIP application	4.2	5.2 commissioning	6-month METI review — must submit by M12
Building permit	4.1	5.1 construction start	Cannot commence construction without it
NTT fiber initiation	4.5 / 4.7	5.7	6–9 month NTT lead time
Gate 3 funding	4.7	5.0 (all)	Phase 3 cannot start without confirmed ¥80M+
Building renovation	5.1	5.5	Data center cannot be fitted before shell complete
Solar + battery	5.2 + 5.3	5.4	EV charging dependent on local power supply

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Rob Oudendijk — YR-Design / Safecast Mitsue, Nara Prefecture, Japan May 2026